

Bellabeat Case Study

Google Data Analytics Professional Certificate – Capstone Project

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ASK — Define the Business Task

Bellabeat aims to better understand how consumers use smart devices in their daily lives. By analyzing publicly available Fitbit fitness tracker data, the goal is to identify user behavior trends and apply these insights to one Bellabeat product to support data-driven marketing strategy decisions.

Key Questions

- What are the main trends in smart device usage?
- How do these trends relate to Bellabeat's target customers?
- How can these insights inform Bellabeat's marketing strategy?

Stakeholders

- Urška Sršen — Cofounder & Chief Creative Officer
- Sando Mur — Cofounder & Executive Team Member
- Bellabeat Marketing Analytics Team
- Bellabeat Executive Team

Business Value

Insights from this analysis will help Bellabeat understand how consumers interact with wellness devices, identify behaviors aligned with its target audience, strengthen product positioning and messaging, and develop more effective, evidence-based marketing campaigns.

PREPARE — Describe the Data Sources

Data source: FitBit Fitness Tracker Dataset (CC0: Public Domain), made available through Mobius on Kaggle. The dataset contains personal fitness tracker data submitted by 30 consenting Fitbit users, including daily activity, sleep, and heart rate records.

Files Used

- **dailyActivity_merged.csv** — daily steps, distance, activity-intensity minutes, calories
- **sleepDay_merged.csv** — daily sleep records (minutes asleep, time in bed)

Data Credibility (ROCCC Summary)

Criterion	Rating	Note
Reliable	Medium	Wearable sensor accuracy varies by device and placement
Original	Medium	Third-party survey participants, not Bellabeat's own users
Comprehensive	Medium	Covers activity + sleep, but small sample, no demographic data

Current	Low	Data collected in 2016
Cited	High	Widely used and referenced in analytics education

Conclusion: the dataset is appropriate for identifying general behavioral trends, but not for large-scale predictions or claims about Bellabeat's specific customer base.

Data Limitations

- Small sample size (33 users with valid activity data, 24 with valid sleep data after cleaning)
- Users may not represent Bellabeat's demographic (no age, gender, or location data available)
- Data is several years old (collected in 2016)
- Sleep data incomplete for some users; heart rate data not used in this analysis

PROCESS — Clean and Prepare the Data

Tools used: Python (pandas, NumPy) for data cleaning, merging, and exploratory analysis; matplotlib and seaborn for visualization. The full, executable code is available in the accompanying Jupyter notebook (see link on the final page).

Data Cleaning Steps

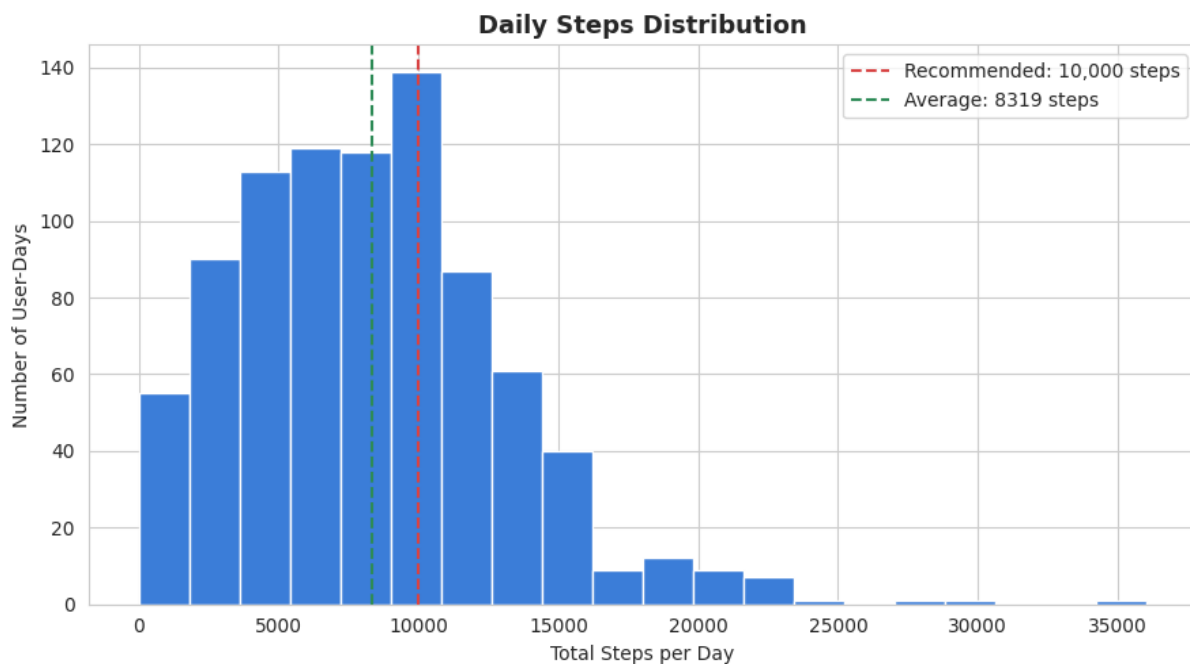
- Checked for and removed duplicate rows (3 duplicate rows found and removed in the sleep dataset)
- Converted date columns in both files to a consistent datetime format
- Removed invalid entries: rows with 0 steps, 0 calories, or 0 minutes asleep, since a value of zero typically indicates the device was not worn that day rather than genuine zero activity
- Merged activity and sleep data on Id and date for the subset of days where both were recorded
- Created calculated fields: TotalActiveMinutes, SedentaryPercentage, SleepEfficiency, DayOfWeek

Validation

- No duplicate rows remain
- No negative or otherwise invalid numeric values
- No missing dates
- Sleep efficiency values fall within a realistic 0–100% range

ANALYZE — Identify Trends and Generate Insights

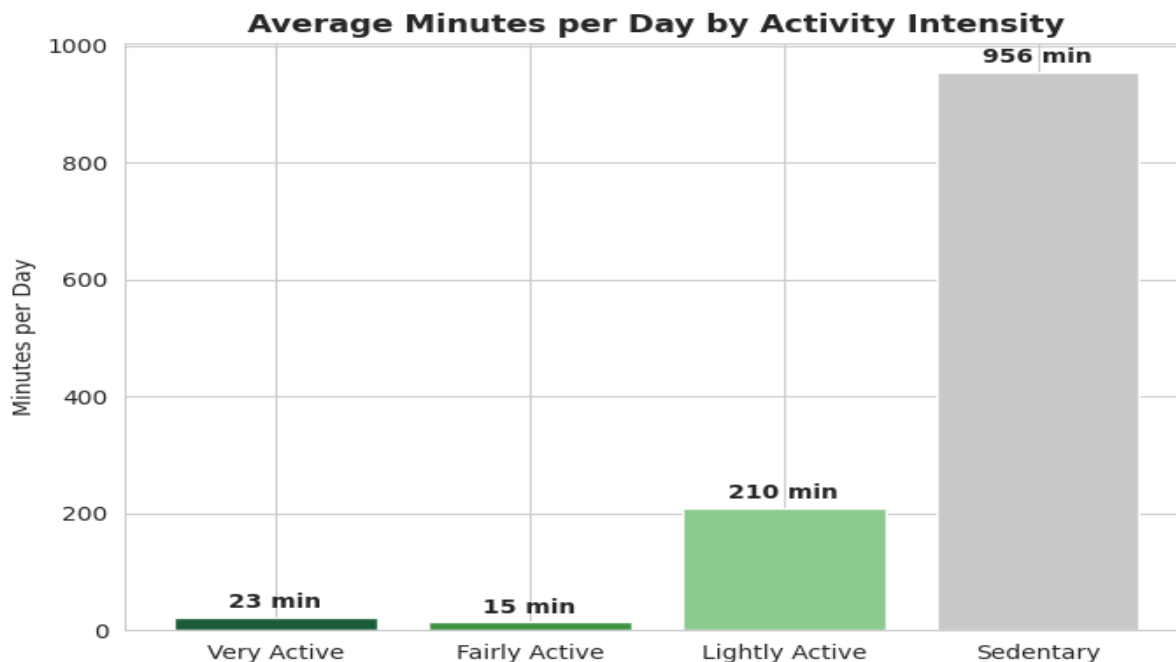
Daily Steps Distribution



Insight: most user-days fall in the 4,000–10,000 step range. The average of roughly 8,300 steps/day sits below the commonly cited 10,000-step benchmark, and only about a third of days reach it — indicating generally low-to-moderate activity levels.

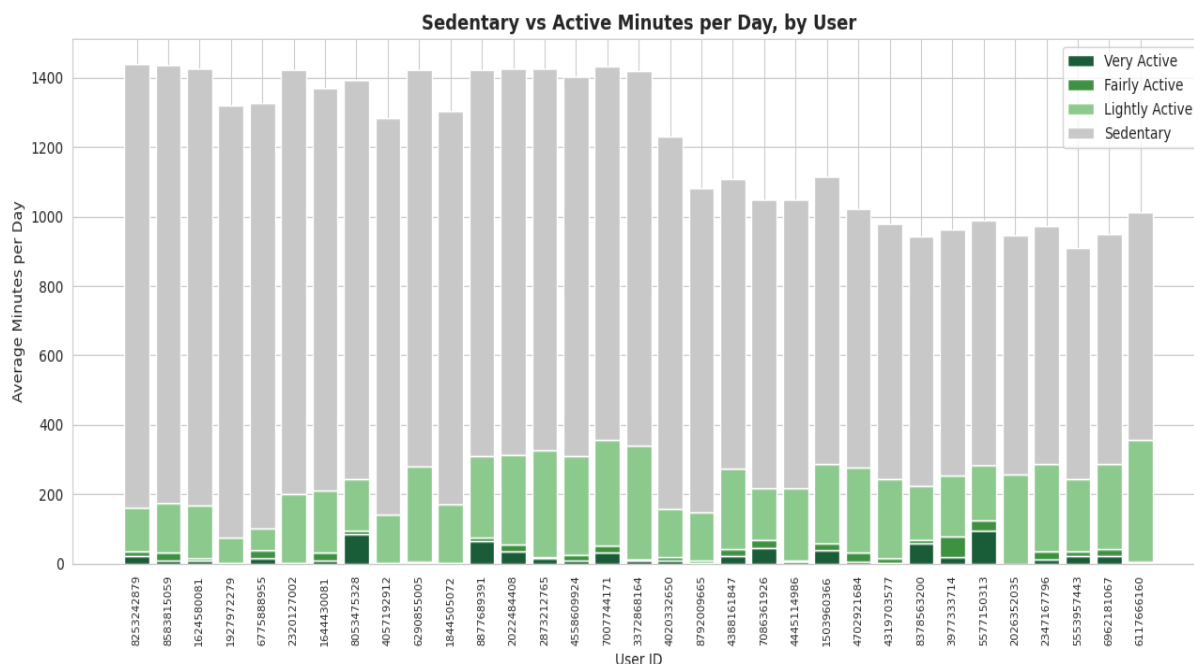
Interpretation: Bellabeat can encourage increased daily movement through reminders and challenges aimed at closing this gap gradually rather than assuming most users are already near the benchmark.

Active Minutes Breakdown by Intensity



Insight: sedentary time dominates the average tracked day, accounting for roughly three-quarters of total minutes. Light activity makes up most of the remaining time; very active minutes are consistently the smallest category.

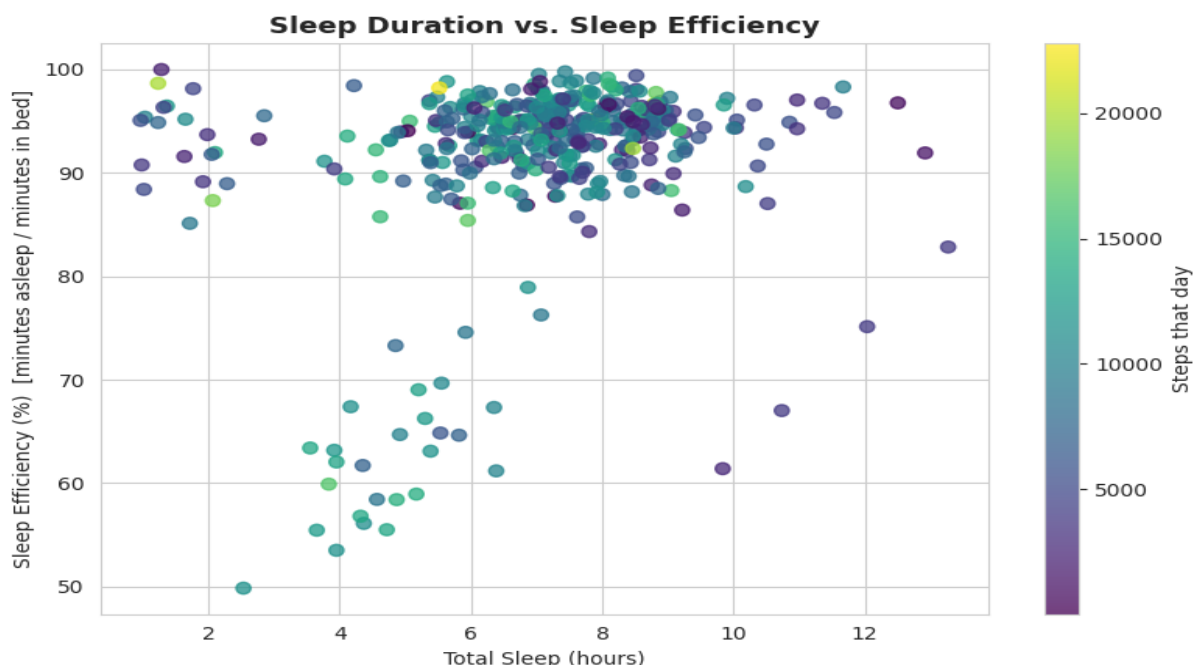
Sedentary vs. Active Minutes by User



Insight: this pattern holds across almost every individual user, not just on average — sedentary time is the largest single component of the day for nearly all 33 users in the sample.

Interpretation: Bellabeat can support users with micro-movement prompts and routine-based nudges rather than positioning its products around high-intensity workouts, since sustained low-intensity movement is where the largest behavior-change opportunity lies.

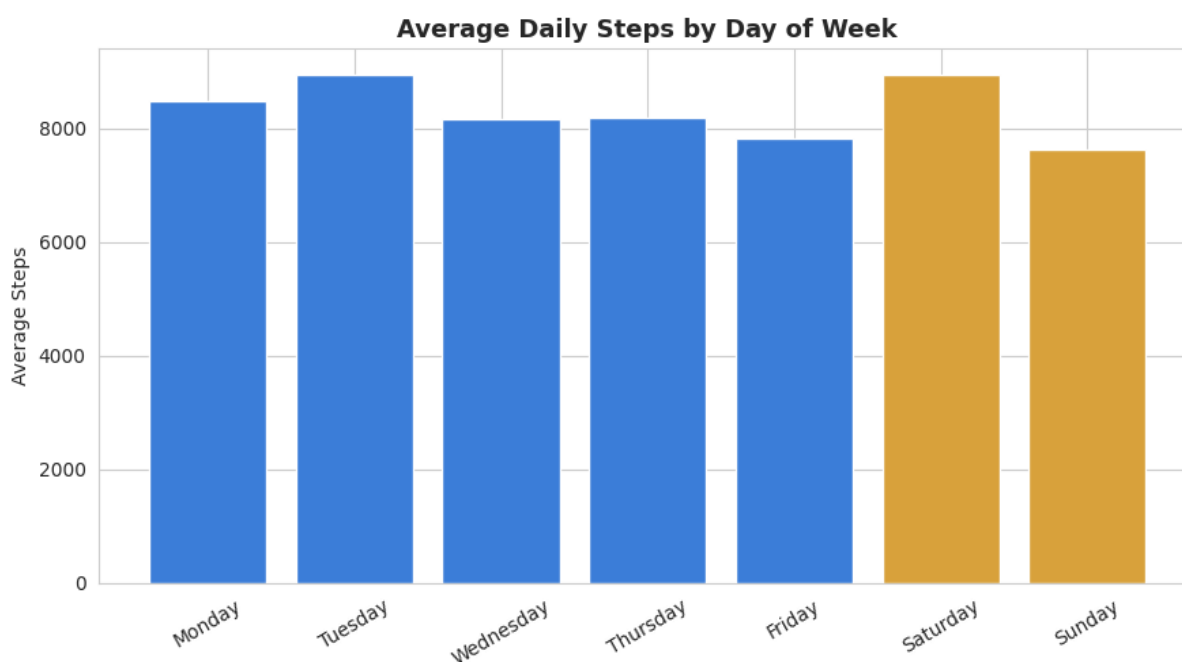
Sleep Duration vs. Sleep Efficiency



Insight: average sleep efficiency is high (~92%) — on nights users are in bed, they are asleep for most of that time. Sleep duration itself, however, is inconsistent across users and nights.

Correction from the original draft: this dataset does **not** show a positive relationship between daytime activity and sleep duration — the measured correlation between daily steps and minutes asleep is weak and slightly negative ($r \approx -0.19$). The claim that "higher activity days correlate with longer sleep" is not supported here and has been removed. The defensible takeaway is that sleep efficiency is a genuine strength to highlight, while sleep duration should not be framed as a downstream effect of daytime activity based on this dataset.

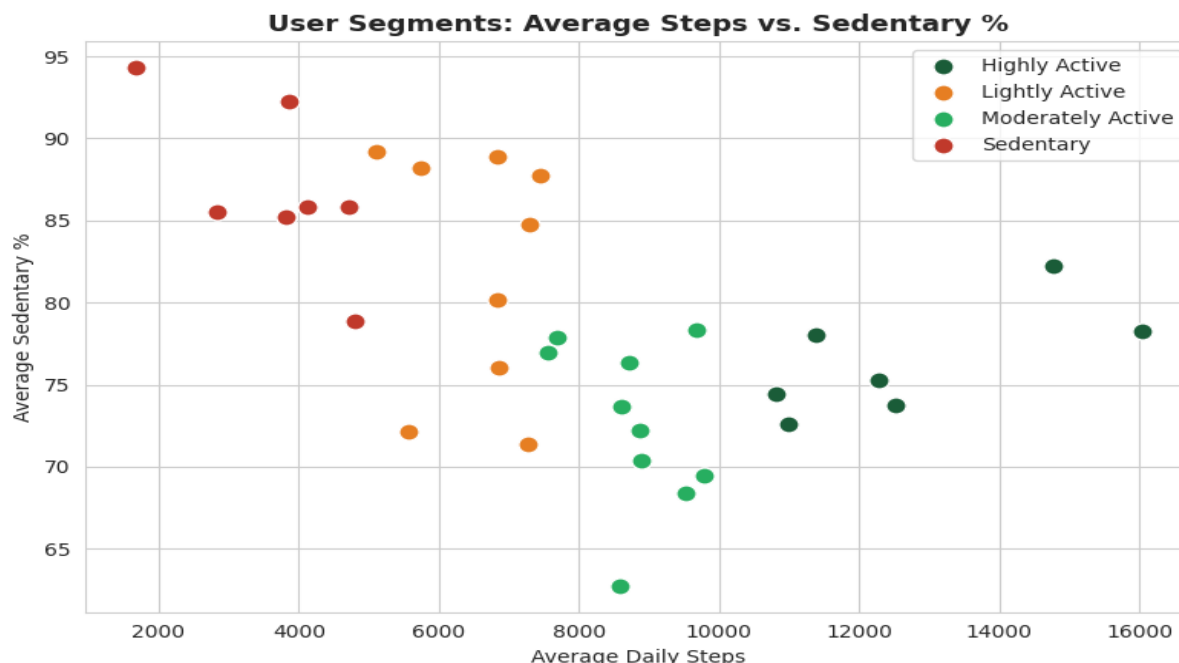
Weekday vs. Weekend Patterns



Insight: average steps are nearly identical on weekdays (~8,330) and weekends (~8,300).

Correction from the original draft: the claim that "weekdays are structured but sedentary, while weekends are more variable" is not supported by this dataset — there is no meaningful weekday/weekend gap in average step counts. Day-to-day variation exists (see chart above), but it does not follow a clean weekday/weekend split, so this framing has been removed rather than asserted.

User Segmentation



Users were grouped into four segments using simple average daily step thresholds: Sedentary (<5,000), Lightly Active (5,000–7,500), Moderately Active (7,500–10,000), and Highly Active (10,000+).

Methodology note: these segments are defined by rule-based thresholds on a single variable (average steps), not by statistical clustering. This is a transparent and reasonable way to segment users for a portfolio project, but it should not be presented as data-driven clustering. A natural next step would be k-means clustering on standardized features (steps, sedentary %, active minutes) to test whether similar groups emerge empirically.

Key Trends Summary

- High sedentary time across nearly all users (~75% of the tracked day)
- Low-to-moderate overall activity levels relative to standard step benchmarks
- Strong sleep efficiency, but inconsistent sleep duration not explained by daytime activity
- No meaningful weekday/weekend difference in step counts in this sample
- Distinct user segments exist and could support differentiated messaging

SHARE — Communicate Findings

Key Insights

- Users spend most of the tracked day sedentary, not just a subset of users
- Overall activity levels are generally low-to-moderate relative to the 10,000-step benchmark
- Sleep efficiency is strong; sleep duration is inconsistent and not linked to daytime activity in this data
- Weekday and weekend activity levels are effectively the same in this sample
- Four distinct behavior-based user segments emerge from simple thresholding

Key Messages for Stakeholders

- High sedentary time signals a real opportunity for movement-guidance features, not just workout tracking
- Low-to-moderate step counts suggest activity-focused features should meet users where they are, rather than assuming most are already close to standard benchmarks
- Sleep-efficiency strength is a safe, evidence-backed marketing angle; a causal link to daily activity is not
- Segment-specific messaging is viable, but should be labeled as threshold-based rather than data-clustered

Executive Summary

Smart device users in this dataset show high sedentary time and low-to-moderate activity levels, with strong sleep efficiency but inconsistent sleep duration. Unlike the initial draft of this analysis, the data does not support a positive activity–sleep-duration relationship or a clear weekday/weekend activity split; both claims were checked against the numbers and corrected. The most defensible opportunity for Bellabeat is to position its products around sustained, low-intensity movement and sleep-efficiency coaching, backed by transparent, verifiable behavioral segments.

ACT — Apply Insights and Recommend Next Steps

Top Recommendations

- 1. Position Bellabeat around sustainable, low-intensity movement.** The data shows this is where most users actually spend their active time — not high-intensity workouts.
- 2. Implement micro-movement, routine-based notifications.** Examples: hourly stand/move prompts, afternoon movement nudges, evening wind-down reminders.
- 3. Market sleep-efficiency insights carefully and accurately.** Efficiency is a genuine, data-backed strength; avoid claiming a link between daily activity and sleep length.
- 4. Use segment-specific campaigns, labeled transparently as threshold-based:**
 - Sedentary → movement challenges and gentle onboarding
 - Lightly / Moderately Active → consistency-building routines
 - Highly Active → performance insights and advanced tracking features

Next Steps for the Business

- Validate these directional findings against Bellabeat's own first-party user data before using them in an actual campaign, since this dataset is third-party, dated, and demographically unverified
- Collaborate with product/UX teams to refine notification timing and content

- Explore k-means clustering as a more rigorous segmentation method
- Expand the analysis using the heart-rate and weight-log files included in the same dataset
- Run A/B tests on wellness-focused messaging before scaling any campaign

Full analysis code, data cleaning steps, and chart-generation code are available in the accompanying Jupyter notebook: **bellabeat_case_study.ipynb** (GitHub / Kaggle).